

BGIS4175 14th Avenue
Markham, Ontario L3R 5R5

REPORT ON

BGIS - Edmonton-HVAC Assessment Floor 5 – 10205 101 Street NW

04/06/2026

File # 226-0489

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1. Introduction

1.01 Scope of Work

1.01.1 This report was prepared at the request of The TD Bank Financial Group and BGIS to provide a high-level due diligence review of the existing HVAC systems serving portions of the 5th floor at Centre Point Place, Edmonton, Alberta. The review was initiated primarily due to occupant comfort complaints within several enclosed perimeter offices located along the west side of the floor plate, along with broader hot/cold concerns reported throughout portions of the tenant space.

- (1) Review accessible existing HVAC systems and observed conditions
- (2) Review of available documentation and background information
- (3) Understand the current system arrangement and known concerns
- (4) Identify potential contributing factors associated with occupant comfort complaints
- (5) Provide high level recommendations and a proposed path forward for further investigation and assessment

1.01.2 Our scope of work included the following:

- (1) Site review of accessible HVAC systems and identified concern areas
- (2) Review of available tenant renovation drawings and limited mechanical documentation
- (3) Discussions with BGIS, landlord operations staff, controls personnel, and onsite contractors
- (4) Review of available information regarding occupancy increases and tenant densification
- (5) High level review of the broader floor HVAC arrangement and observed operational concerns
- (6) Preparation of a due diligence report outlining observations, findings, and recommended next steps

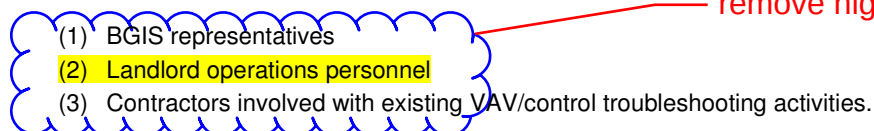
1.01.3 This review was not intended to constitute:

- (1) A full floor mechanical as-built verification
- (2) A detailed engineering design study
- (3) A TAB exercise
- (4) A full HVAC zoning redesign
- (5) Or a comprehensive heating/cooling load analysis of the entire floor plate

1.02 Methodology

1.02.1 A limited visual review of accessible HVAC systems and tenant areas was completed during a single site visit.

1.02.2 Discussions were held onsite with:

- 
- (1) BGIS representatives
 - (2) Landlord operations personnel
 - (3) Contractors involved with existing VAV/control troubleshooting activities.

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1.02.3 Available renovation drawings and limited mechanical documentation were reviewed. Documentation available at the time of review primarily consisted of partial tenant renovation packages from approximately 2016, 2017, and 2021.

- 1.02.4 The only broader floor mechanical drawings identified onsite during the review were construction issue drawings dated August 25, 2008. These drawings were reportedly located onsite by the tenant and were not available within landlord archives at the time of review.
- 1.02.5 No detailed airflow measurements, TAB activities, duct traverses, heating/cooling load calculations, ventilation calculations, or ceiling investigations were completed as part of this review.

1.03 Limitations

- 1.03.1 This report has been prepared solely for the use and benefit of The TD Bank Financial Group and BGIS and may not be relied upon by others without written consent from H.H. Angus & Associates Limited.
- 1.03.2 The review did not include:
- (1) Full ductwork tracing
 - (2) Full VAV mapping
 - (3) Detailed controls verification
 - (4) Airflow measurements
 - (5) TAB Services
 - (6) Verification of concealed conditions
 - (7) Or detailed engineering calculations
- 1.03.3 Given the apparent age of the systems, multiple generations of tenant renovations, legacy controls infrastructure, undocumented modifications, and limited available as-built documentation, additional investigation will be required prior to developing final engineering conclusions or major HVAC upgrade recommendations.

2. Existing Building and HVAC System Overview

2.01 Building Background

- 2.01.1 The building was originally constructed as the Woodward's department store in 1972. The upper floors (floors 4 & 5) have since been converted into office tenant/call centre occupancy, while lower floors (floors 1-3) remain retail occupancies.
- 2.01.2 The reviewed tenant space is located primarily on the 5th floor and is currently utilized as TD office/call centre space.
- 2.01.3 Based on discussions onsite, the floor has undergone multiple tenant renovations and workstation reconfigurations over time, including densification associated with increased occupant counts and revised furniture layouts.
- 2.01.4 Discussions onsite suggested that current occupancy may now approach approximately 400 to 460 occupants on the floor plate. Occupancy counts and operating conditions should be confirmed as part of future investigation activities.

2.02 Heating, Ventilation, and Air Conditioning (HVAC)

- 2.02.1 Existing System Observations

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- (1) Cooling and ventilation to the floor are provided by older central air handling/VAV distribution systems serving the tenant floor plate.



Figure 1 –Mechanical Room AHU compartment

- (2) Two primary air handling systems reportedly serve the building floor plates:
- One system serving the north side
 - One system serving the south side

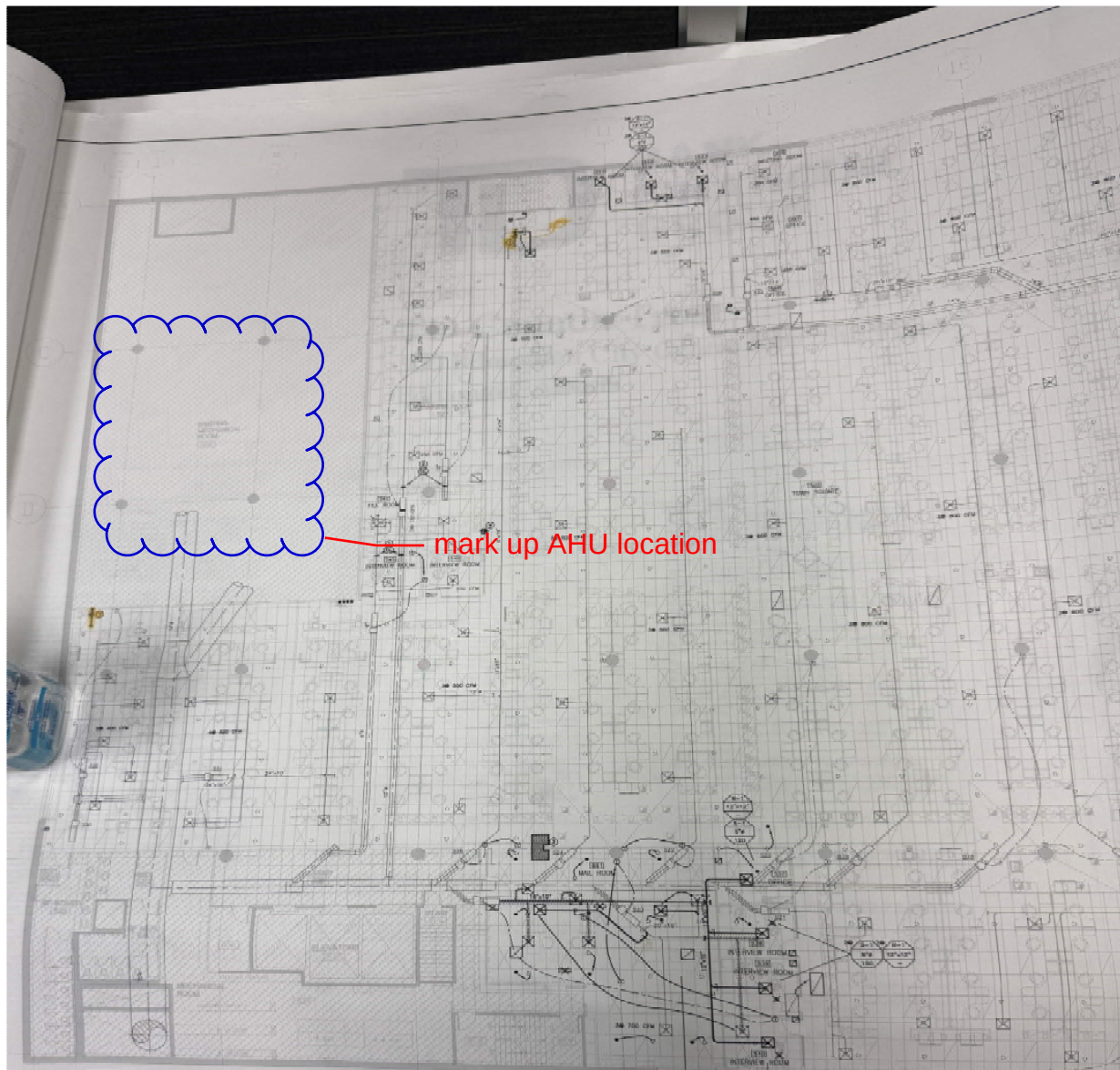


Figure 2 – North Side Air Handling System



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The systems reportedly serve multiple levels of the building, including both office and retail occupancies.

- (3) The office floors are served primarily by VAV terminal units located within the ceiling plenum.
- (4) The building is served by a central chilled water plant reportedly consisting of approximately three 1000-ton chillers serving both office towers and portions of the retail areas.
- (5) Landlord personnel advised that portions of the existing AHU systems are currently undergoing staged upgrades and repair planning, including cooling coil replacement work and installation of isolation valves to facilitate future maintenance activities.
- (6) The HVAC systems serving the floor have evolved over many years through multiple partial tenant renovations and occupancy changes, while portions of the underlying HVAC infrastructure and zoning strategy remain older legacy systems.
- (7) Multiple generations of thermostats, sensors, VAV controllers, and controls hardware were observed on the 5th floor.



Figure 4 – Thermostat #1 (Generation 1)



Figure 5 – Thermostat #2 (Generation 2)



Figure 6 – Thermostat #3 (Generation 3 – Newest upgrade)

- (8) Several older thermostats reportedly remain onsite despite no longer being actively utilized. This raises the possibility that portions of the existing systems may include abandoned controls devices, abandoned ductwork, undocumented zoning changes, or legacy installations that are no longer fully understood.



Figure 7 – Older Inactive Thermostat (Generation 1)

- (9) Several VAV controllers were reportedly offline or non-reporting at the time of review.
- (10) Discussions onsite indicated that thermostat locations may not align well with current room usage or occupancy patterns. Several enclosed rooms are controlled by thermostats located within adjacent open office areas.
- (11) Based on discussions onsite, portions of the floor may contain multiple enclosed spaces served by single VAV terminal units, limiting the ability to independently control occupant comfort conditions within enclosed perimeter spaces.
- (12) The reviewed floor has undergone substantial workstation densification, and occupancy increases over time without corresponding comprehensive HVAC redesign, re-zoning, or modernization of the overall floor control strategy.
- (13) At the time of review, insufficient information was available to determine:
- Actual outdoor air quantities serving the floor
 - Actual delivered airflow quantities
 - Whether current systems are operating as originally intended
 - Whether the systems are currently balanced properly
 - Whether the floor is operating beyond original design assumptions
 - Whether the observed issues are primarily operational, zoning related, maintenance related, or capacity related.
- (14) The existing base building systems may have originally been designed around significantly different occupancy assumptions associated with the original department store configuration and/or earlier office layouts.
- (15) The base building systems can provide cooling and ventilation only within the limits of their original design capacities. Increases in tenant heat loads or occupant densities may ultimately require supplemental cooling strategies beyond existing base building capacity.
- (16) Perimeter heating is provided throughout portions of the floor plate. The existing zoning arrangement, control strategy, and relationship between perimeter heating and interior cooling systems were not verified during this review and should be assessed as part of any future HVAC investigation.

2.02.2 Documents and Information Reviewed

The following information was made available during the review:

- (1) Base building architectural, mechanical and structural drawings provided by Edmonton City Centre.
- (2) Tenant renovation drawing packages dated approximately 2016, 2017 and 2021.
- (3) Available operation and maintenance documentation associated with previous tenant renovations.
- (4) Building Automation System (BAS) graphics provided for the 5th Floor VAV North and South systems only.
- (5) Site discussions with BGIS, Edmonton City Centre operations staff, controls contractors and maintenance personnel.
- (6) Historical HVAC service call information provided by Edmonton City Centre.
- (7) Physical construction drawings dated August 25, 2008 located onsite within the tenant space.

2.02.3 Reliability of Available Documentation

Complete current 5th floor configuration was not available. The only known floor wide HVAC drawing package identified during the review was a physical construction drawing set dated August 25, 2008 located onsite within the tenant space.

2.02.4 Recent and Planned Base Building Initiatives

Based on discussions with Edmonton City Centre operations personnel, several base building initiatives are either underway or planned.

- (1) Five identified VAV controller replacements have been completed, (VAV-504, VAV-508, VAV-520 and VAV- 532). To date VAV-502 remains outstanding due to access limitations.
- (2) Edmonton City Centre has approved replacement of the cooling coils serving both north and south air handling systems. Replacement is anticipated toward late summer or early fall pending procurement and contractor scheduling.
- (3) Additional future air handling unit upgrades under consideration include:
 - a. Supply fan replacement
 - b. Return fan replacement
 - c. Damper and actuator replacement
 - d. Humidification upgrades
 - e. Building automation system upgrades and modernization
- (4) These initiatives were noted for general understanding only and were not evaluated as part of this review.

2.02.5 BAS

BAS graphics were reviewed for both the north and south air handling systems and associated VAV terminal units. The graphics indicate the floor is served by multiple VAV zones connected to separate north and south air handling systems. Review of the graphics identified ongoing controller replacement activities and highlighted the presence of multiple control generations throughout the floor plate. The BAS graphics reviewed do not provide sufficient information to fully understand current zoning relationships, thermostat assignments, airflow quantities, or the impacts of subsequent tenant renovations.

2.03 North Perimeter Office Comfort Concerns

2.03.1 Observations

- (1) The original primary concern identified by BGIS and the tenant involved several enclosed perimeter offices located along the north side of the floor plate.
- (2) Based on site observations and discussions onsite, the affected perimeter offices are served by a single VAV terminal unit shared with adjacent interior/open office areas.
- (3) The thermostat associated with this zone is located within an interior/open office area rather than within the perimeter offices themselves.
- (4) The existing zoning arrangement may result in:
 - Poor perimeter temperature control
 - Simultaneous overcooling/undercooling conditions
 - Inability to independently control perimeter solar/load impacts
 - Inconsistent occupant comfort conditions between enclosed offices and adjacent open office areas
- (5) The existing arrangement is inconsistent with modern perimeter zoning typically utilized for dense office occupancy layouts.

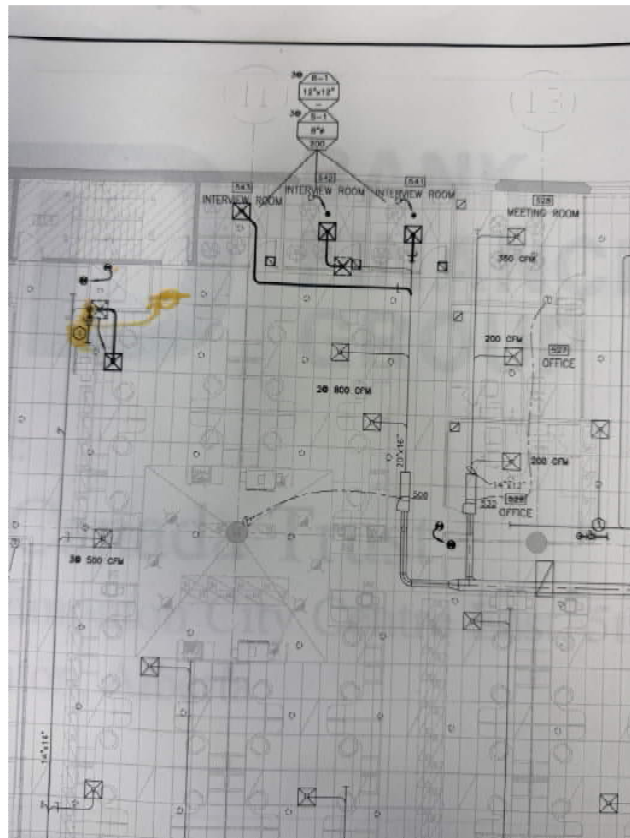


Figure 8 – Perimeter Offices Served By An Interior Thermostat



Figure 9 – VAV Serving Perimeter Offices from Figure 8

3. Recommendations

3.01 North Perimeter Offices

3.01.1 It is recommended that the north perimeter offices be separated from the existing shared VAV zone and provided with dedicated perimeter zoning control.

3.01.2 This would likely include:

- Installation of a new dedicated VAV terminal unit
- Installation of a dedicated perimeter thermostat
- Rebalancing/downsizing of the existing shared VAV airflow quantities as required
- Review of perimeter solar loading conditions and blind operation practices

3.01.3 Additional field verification would be required prior to final design.

3.02 Broader Floor HVAC Investigation

3.02.1 Documentation and Existing System Review

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3.02.2 The first recommended step is development of a more complete understanding of the existing systems and floor conditions. This should include:

- (1) Collection of all available landlord and tenant drawings
- (2) Collection of BAS graphics and controls information
- (3) Collection of historical TAB reports
- (4) Confirmation of occupancy counts and workstation densities
- (5) Review of historical maintenance/service call records
- (6) Review of current zoning and controls arrangements

3.02.3 Existing Conditions Verification

Following the documentation review phase, field verification by a qualified TAB and controls contractor is recommended, including:

- (1) duct tracing
- (2) VAV verification
- (3) thermostat mapping
- (4) controls verification
- (5) airflow measurements
- (6) TAB review
- (7) Outdoor air verification
- (8) Review and verification of perimeter heating zoning and control strategy
- (9) Verification of active versus abandoned thermostats, sensors, and control devices
- (10) Verification of active versus abandoned ductwork and HVAC infrastructure

3.02.4 HVAC Load and Ventilation Review

Once sufficient field information is obtained, updated engineering calculations should be completed, including:

- (1) Occupancy verification
- (2) Heating load calculations
- (3) Cooling load calculations
- (4) Ventilation calculations
- (5) Outdoor air
- (6) Comparison of calculated ventilation requirements versus available outdoor air delivery
- (7) Comparison of calculated heating and cooling loads versus available system capacity

These calculations should then be compared against:

- (8) Available system airflow
- (9) Measured air quantities
- (10) Available cooling capacity
- (11) Existing zoning arrangements

3.02.5 Supplemental Cooling Considerations

If future investigation determines that the existing base building systems are insufficient for current tenant occupancy and cooling requirements, supplemental cooling strategies may need to be evaluated.

Given the tenant space is located on the upper floor of the building, discussions onsite indicated that rooftop mounted supplemental equipment may be feasible subject to landlord review and approval. Prior to

consideration of supplemental cooling solutions, the existing HVAC systems should first be verified, balanced, and evaluated to determine whether the observed issues are related to capacity limitations, zoning deficiencies, control issues, maintenance concerns, or a combination thereof.

Potential future options could include:

- (1) Rooftop mounted supplemental HVAC equipment
- (2) Supplemental DX cooling systems
- (3) Fan coil systems
- (4) Localized supplemental cooling strategies
- (5) Additional perimeter cooling solutions

These options would require further engineering review and coordination with the landlord and tenant teams.

3.02.6 Recommended Investigation Sequence

- (1) Review available documentation including the 2008 construction drawings, subsequent tenant renovation drawings, BAS graphics, maintenance records, occupancy information, and historical service records.
- (2) Engage a qualified TAB contractor and controls contractor to complete field verification activities, including:
 - VAV box identification and operation
 - Thermostat and controls verification
 - Perimeter heating zoning verification
 - Airflow measurements
 - Outdoor air assessment
 - System balancing review
 - Identification of active versus abandoned HVAC infrastructure
- (3) Utilize the available drawings and field verification information to develop an updated understanding of the existing HVAC systems serving the floor.
- (4) Complete ventilation, heating, and cooling load calculations based on the current floor layout, occupancy levels, and operational requirements.
- (5) Compare calculated requirements against measured system performance and available HVAC capacity.
- (6) Develop HVAC improvement options based on verified field conditions and engineering calculations.
- (7) Implement selected improvements and complete final TAB verification following any modifications.

4. Conclusions

4.01 Mechanical Conclusion

- 4.01.1 Based on the limited review completed, the current HVAC concerns are not isolated solely to the originally identified perimeter offices. Discussions onsite and available information suggest broader system, zoning, occupancy, controls, and documentation related concerns may exist throughout portions of the floor plate.
- 4.01.2 A significant contributing factor may be the increase in occupant density, workstation count, and space reconfiguration over time without corresponding comprehensive HVAC redesign, re-zoning, controls modernization, or verification of ventilation and cooling requirements.

- 4.01.3 At the time of review, insufficient information was available to conclusively determine whether the existing base building HVAC systems are fundamentally undersized for the current occupancy conditions, or whether operational/control/zoning deficiencies are the primary contributors to the observed comfort concerns. This determination will require verification of actual airflow quantities, outdoor air delivery, occupancy levels, zoning arrangements, and HVAC system performance.
- 4.01.4 Additional investigation, controls review, airflow verification, and system mapping activities are recommended prior to implementation of any major capital HVAC modifications.
- 4.01.5 The north perimeter office concerns are more localized in nature and likely attributable to the existing zoning arrangement and thermostat placement associated with the current shared VAV configuration.
- 4.01.6 A structured investigation approach is recommended consisting of documentation review, TAB and controls verification, HVAC system assessment, engineering calculations, and subsequent development of improvement options based on verified field conditions.
- 4.01.7 Available documentation is incomplete and, in some cases, may not accurately reflect current field conditions. Development of a reliable floor wide HVAC strategy will require field verification of system zoning, controls, ductwork distribution, and airflow delivery prior to implementation of major modifications.

Report prepared by: Ash Parmar